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The impact of multi-disciplinary intestinal rehabilitation programs on the outcome of pediatric patients with intestinal failure: A systematic review and meta-analysis

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Abstract

Background: Pediatric intestinal failure (IF) is a complex clinical problem requiring coordinated multi-disciplinary care. Our objective was to review the evidence for the benefit of intestinal rehabilitation programs (IRP) in pediatric IF patients.

Methods: A systematic review was performed on Medline (1950–2012), Pubmed (1966–2012), and Embase (1980–2012) conference proceedings and trial registries. The terms short bowel syndrome, intestinal rehabilitation, intestinal failure, patient care teams, and multi-disciplinary teams were used. Fifteen independent studies were included. Three studies that were cohort studies, including a comparison group, were included in a meta-analysis.

Results: Compared to historical controls (n = 103), implementation of an IRP (n = 130) resulted in a reduction in septic episodes (0.3 vs. 0.5 event/month; p = 0.01) and an increase in overall patient survival (22% to 42%). Non-significant improvements were seen in weaning from PN (RR = 1.05, 0.88–1.25, p = 0.62), incidence of IFALD (RR = 0.2, 0–17.25, p = 0.48), and relative risk of liver transplantation (3.99, 0.75–21.3, p = 0.11). Other outcomes reported included a reduction in calories from parenteral nutrition (100% to 32%–56%), earlier surgical/transplant evaluation, and improved coordination of patient care.

Conclusion: For pediatric IF patients, IRPs are associated with reduced morbidity and mortality. Standardized clinical practice guidelines are necessary to provide uniform patient care and outcome assessment.

Pediatric intestinal failure is a complex clinical problem that is associated with high patient mortality and morbidity. Intestinal failure results from short bowel syndrome, mucosal enteropathy or intestinal dysmotility, all of which result in inappropriate nutrient absorption to promote normal growth and development.

The survival of pediatric patients with intestinal failure has traditionally been dismal, with greater than 40%

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mortality. The common causes of death in this population have been related to sepsis or end stage liver disease from intestinal failure associated liver disease (IFALD) [1]. Developments in parenteral nutrition and its administration, such as alternate sources of lipids, increased awareness of infection prevention and home nutrition programs, have resulted in improved patient outcomes [2–7]. Effectively caring for these patients and their families is a complex task that is best completed in a coordinated, multi-disciplinary setting [1]. The benefits of a multi-disciplinary approach include the amalgamation of expert care, a holistic approach to patients, stream-lined diagnosis and treatment, strengthened communication with both, families and the care teams, consistent follow up and continuity of patient care. To that end, several pediatric institutions across North America and Europe have developed comprehensive, multi-disciplinary teams for the treatment and rehabilitation of children with intestinal failure [2–4,8–20]. The objective of this paper was to review the evidence for the benefit of multi-disciplinary teams in the management and outcomes of pediatric patients with intestinal failure.

1. Methods

This systematic review and meta-analysis is reported according to the PRISMA-statement [21,22]. Methods of the analysis and inclusion criteria were specified in advance, but no protocol was published, nor has the study been registered.

1.1. Literature search

A search of the medical literature was conducted using Medline (1948–2012), Pubmed (1966–2012) and Embase (1980–2012) databases in March 2012. The Conference Proceeding Citation Index of The Web of Science, Scopus, Cochrane Central Register of Controlled Trials (CENTRAL), CinicalTrials.gov and WHO trials registry were searched in March 2012. Abstract lists from the annual meetings of major pediatric surgery and selected pediatric medical associations from the last three years (2008–2010) were scanned by one author (JS): Canadian Association of Paediatric Surgeons, American Pediatric Surgical Association, European Paediatric Surgeons' Association, British Association of Paediatric Surgeons, and American Academy of Pediatrics. The search terms *short bowel syndrome*, *intestinal rehabilitation*, *intestinal failure*, *patient care teams* and *multi-disciplinary teams* were used individually and in combination. The search was not restricted by language or time of publication. Pediatric patients with the diagnosis of intestinal failure were included. No restrictions regarding the characteristics of IRP, definition of intestinal failure and age at diagnosis were applied. The abstracts of all identified papers were reviewed by two independent reviewers (JS, CB). Discrepancies were reviewed by the senior author (PW). Papers were excluded if they did not include an outcome measure of the program. An additional hand search was conducted of the reference lists of all eligible papers.

1.2. Meta-analysis

All study designs comparing IRP to traditional management of pediatric intestinal failure were included in a meta-analysis. No restrictions regarding language, publication date or state, were applied. In-vitro and animal model studies, as well as, case series and other non-comparative studies were excluded. The primary outcome measure was disease-specific mortality (defined as mortality from liver failure or sepsis) and the main secondary outcome was overall mortality. Additional secondary outcomes were exploratory and included weaning from parenteral nutrition, development of liver failure, and rate of transplantation. Studies that reported on the primary outcome were included.

Two authors (CO, JS) extracted the data in an un-blinded fashion, but independently into an electronic datasheet. Disagreements were solved by consultation of the senior author (PW). No authors were contacted for missing variables. Data abstracted from each participating study included: study characteristics (design, sample size, publication year, study period, population, follow-up), patient characteristics (birth weight, gestational age, diagnosis), surgical outcomes (residual small bowel length, preservation of the ileocecal valve), clinical course (days of parenteral nutrition, septic episodes, development of cholestasis and liver failure, peak direct bilirubin), outcome (weaning from parenteral nutrition, disease-specific mortality, overall mortality) and therapeutic options (IRP components, utilization of novel lipid management strategies [omega-3 intravenous lipid emulsions, lipid minimization], serial transverse enteroplasty, transplantation).

The quality of included studies was assessed using the Newcastle–Ottawa scale [23]. Two authors (CO, JS) independently assessed studies using eight criteria: representativeness of exposed group, selection of control group, ascertainment of exposure, demonstration of lack of outcome at the start of the study, description of excluded patients, assessment of outcome, adequacy of follow-up time and comparability of cohorts. Due to the low number of included studies, we did not test for reporting bias using graphical or statistical methods.

1.3. Statistical analysis

Summary statistics include means with standard deviation and frequencies with proportions where appropriate. Meta-analysis was performed on Review Manager (RevMan), [Version 5.0. Copenhagen: The Nordic Cochrane Centre, The Cochrane Collaboration, 2008]. Appropriate results from individual studies were quantitatively pooled in meta-analyses using fixed- or random-effects models, depending on heterogeneity. Effect sizes were measured primarily in relative risk (RR) and mean difference (MD). Sensitivity analyses were pre-specified. A one-way sensitivity analysis was performed for exclusion of each study at a time. The Q-test was computed using a fixed-effects model to assess the presence of heterogeneity. If significant at an alpha-level of 0.1, heterogeneity was quantified using the I^2-value. If $I^2 < 50\%$, the fixed-effects model was used and if $I^2 \geq 50\%$, the random-effects model was applied [24–27]. For the purpose of the meta-analysis, the intervention or exposure was the intestinal rehabilitation team itself and encompasses all nutritional, pharmacologic and surgical therapies utilized. An alpha value <0.05 was considered significant and 95% confidence intervals (95%-CI) were also included as a measure of precision.

2. Results

2.1. Literature search

Based on the literature search, 127 papers were identified, of which 15 met inclusion criteria, reflecting the outcomes from 13 independent programs [2–4,8–20]. Three of the studies were retrospective cohort studies that compared the outcomes among patients before and after the implementation of a multi-disciplinary program [2–4]. The remainder of the studies were descriptive case series (Fig. 1). Among the studies, a variety of patient outcomes were reported including mortality, septic episodes, bilirubin levels/IFALD, surgical intervention and rates of transplantation. Studies included in the review are summarized in Table 1.

2.2. Meta-analysis

Three studies were included in the meta-analysis (Table 2). The assessment of quality revealed 6 of 8 criteria were met. “Demonstration of absence of outcome of interest at start of study” and “description of excluded patients” were not applicable due to the non-experimental design utilized in these 3 included studies.

The primary outcome of interest was disease specific mortality (death from liver failure and sepsis). Individual and pooled data on the disease-specific mortality are presented in Fig. 2A. Individual and pooled data on secondary outcomes are presented in Fig. 2B–E. Results of the meta-analysis are included in the summary of the systematic review below.

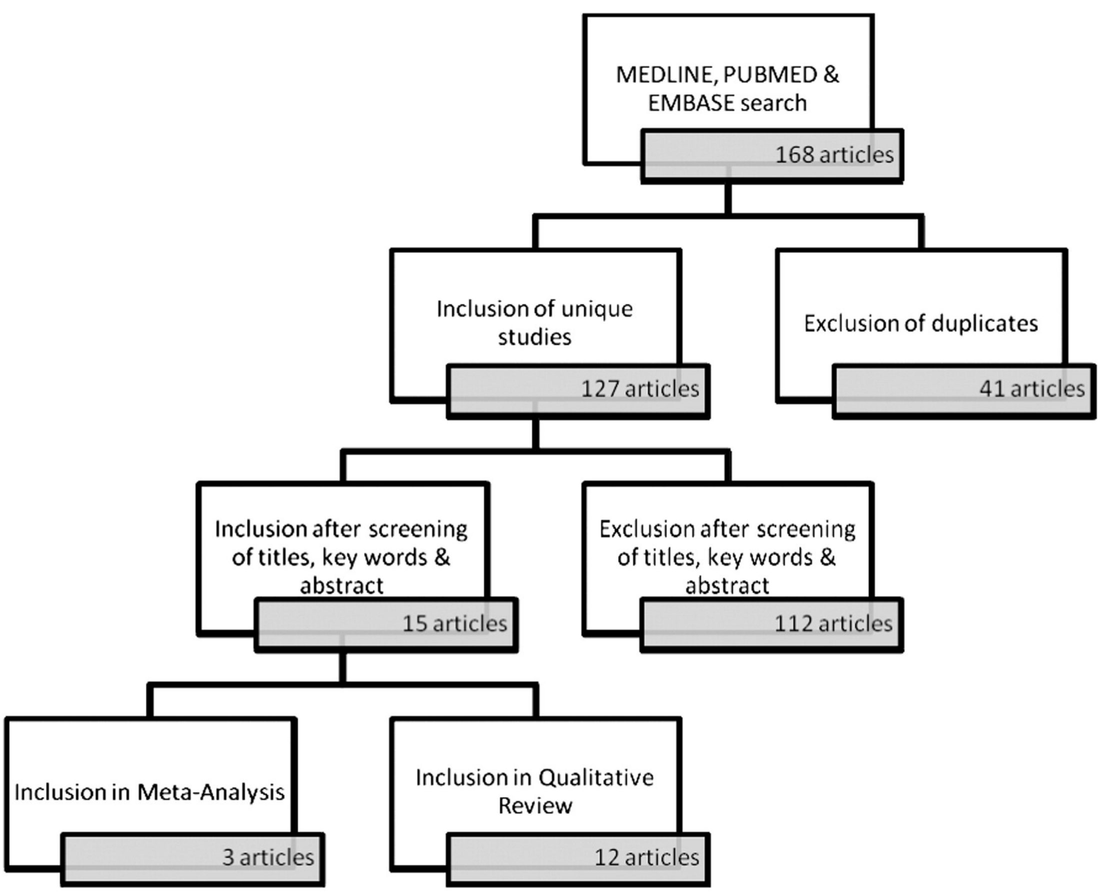

Fig. 1 Systematic Review. The literature search resulted in 15 articles selected for inclusion. After assessment of the full text, three studies were included for meta-analysis.

Table 1 Studies included in the systematic review.

Study	N	Design	Follow Up (months)	Enteral Autonomy		Sepsis Rates (Episodes / month)	IFALD		Transplant	Bowel Lengthening Procedure	Mortality
				Full Feeds N (%)	% decrease in PN Needs		Incidence N (%)	Normalization of Bilirubin	N (%)	N (%)	N (%)
Fishbein et al. 2002.	20	Case Series	14	10 (50)	NR	NR	14 (87.5)	NR	9 (45)	NR	5 (25)
Torres et al. 2007.	51	Case Series	NR	31 (60.8)	68%	NR	36 (70.6)	29 (80.6)	10 (19.6)	29 (56.8)	5 (9.8)
Diamond et al. 2007.	54	Retrospective Comparison	16	35 (64.8)	NR	0.3	43 (79.6)	NR	4 (7.4)	14 (26)	18 (33.3)
Modi et al. 2008.	54	Retrospective Comparison	13.4	36 (66.7)	31 %	NR	35 (64.8)	NR	5 (9.3)	5 (9.3)	6 (11.1)
Nucci et al. 2008.	389	Case Series	24	141 (36.2)	32 %	NR	NR	NR	119 (30.6)	25 (6.4)	109 (28)
Olieman et al. 2010.	10	Cost Analysis	18	6 (60)	NR	0.4	8 (80)	NR	NR	NR	0 (0)
Sigalet et al. 2009	22	Retrospective Comparison	15.4	18 (81.8)	NR	NR	0 (0)	NR	0 (0)	4 (18.2)	0 (0)
Cowles et al. 2010.	93	Case Series	12	NR	NR	NR	76 (82)	57 (75)	15 (16.1)	NR	14 (15.1)
Javid et al. 2010.	49	Case Series	14	22 (45)	69 %	NR	NR	NR	3 (6.1)	13 (26.5)	6 (12.2)
Guarino et al. 2003.	109	Case Series	50	46 (42.2)	NR	NR	NR	NR	4 (3.7)	NR	7 (6.4)
Hess et al. 2011	171	Case Series	24	114 (169)	NR	NR	90 (53.6)	NR	NR	12 (7.36)	21 (12.3)
Colomb et al. 2007.	302	Case Series	NR	163 (54)	NR	NR	213 (71)	NR	13 (4.3)	NR	48 (16)
Gupte et al. 2007.	152	Case Series	NR	NR	NR	NR	NR	NR	32 (21.1)	4 (2.6)	49 (32.2)
Khalil et al. 2012.	27	Case Series	NR	21 (77.8)	NR	NR	NR	NR	0 (00	19 (70.4)	2 (7.4)

Table 2 Studies included in meta-analysis.

		Diamond et al., 2007		Modi et al., 2008		Sigalet et al., 2009	
		IRP group	Control	IRP group	Control	IRP group	Control
Study features	N, time frame	54, 2003–2005	40, 1997–1999	54, 1999–2006	30, 1986–1998	22, 2006–2009	33, 1998–2006
	Population	Patients <6 months with intestinal length <25th %ile OR PN >42 days post-op		Pediatric patients with severe SBS: PN ≥90 days		Pediatric patients with laparotomy AND <40 cm of intestinal length OR PN >60 days post-op	
	Follow-up (days)	489 ± 48	591 ± 68	403 (256–1333)[a]	NR	462 ± 240	2250 ± 480
Patient characteristics	Birth weight (g)	2253 ± 144	1660 ± 148	NR	NR	2340 ± 760	2480 ± 600
	Gestational age (weeks)	33.5 ± 0.7	30.7 ± 0.7	34 (IQR 28–37)	NR	37.1 ± 5.9	33.7 ± 53.4
	Most frequent etiologies	NEC (14), Gastroschisis (5), Mec. Ileus (8)	NEC (19), Gastroschisis (12), Atresia (10)	NEC (16), Atresia (12), Gastroschisis (10)	NEC (13), Atresia (9), Gastroschisis (5)	Gastroschisis (15), Atresia (6), NEC (3)	NEC (11), Gastroschisis (10), Atresia (7)
Anatomy	Bowel length (cm)	92 ± 6	73 ± 7	70 ± 36	83 ± 67	87 ± 47	59 ± 33
	ICV (%)	46 (85.2)	28 (70)	24 (44)	17 (56)	16 (73)	26 (79)
Clinical course	Days of PN	71 (50–116)[b]	76 (46–118)[b]	475	606	NR	NR
	Septic episodes (n/month)	0.3 (0–0.6)[a]	0.5 (0.3–1)[a]	NR	NR	NR	NR
	Cholestasis[c]	43 (79.6)	25 (62.5)	NR	NR	NR	NR
	Liver failure (%)	13 (24.1)[d]	10 (25)[d]	NR	NR	0 (0)[e]	24 (72.7)[e]
	Peak direct bilirubin (μmol/L)	NR	NR	138.5 ± 135.1	153.9 ± 126.6	8 ± 2[f]	112 ± 34[f]
Outcome	Wean PN (%)	35 (64.8)	25 (62.5)	36 (67)	20 (67)	18 (82)	24 (73)
	Disease-specific survival (%)	42 (77.8)	28 (70)	49 (90.74)	22 (73.3)	22 (100)	24 (72.73)
	Overall survival (%)	36 (66.7)	25 (62.5)	48 (88.9)	21 (70)	22 (100)	24 (72.73)
Therapeutic options	Omega-3 fatty acids (%)	0 (0)	0 (0)	14 (46.67)	0 (0)	6 (27.3)	0 (0)
	Fat minimization (%)	0 (0)	0 (0)	0 (0)	0 (0)	8 (36.36)	2 (6.01)
	STEP (%)	14 (25.9)	0 (0)	5 (9.3)	0 (0)	4 (18.2)	0 (0)
	Transplants (%)	4 (7.41)	0 (0)	5 (9.26)	1 (3.33)	0 (0)	0 (0)

Abbreviations: NR not reported, ± standard deviation, [a]median and interquartile range, NEC necrotizing enterocolitis, ICV ileo-cecal valve, PN parenteral nutrition, [b]days to full fees (excl. patients who did not achieve full feeds), [c]direct bilirubin > 50 μmol/L for >2 weeks and not associated with sepsis, [d]direct bilirubin >200 μmol/L for >2 weeks and not associated with sepsis, INR >1.5, albumin <20 g/L, <1000 thrombocytes, portal hypertension, or bridging fibrosis in biopsy, [e]direct bilirubin constantly >100μmol/L, [f]average direct bilirubin after 3 months of PN. All values indicated as means unless otherwise specified.

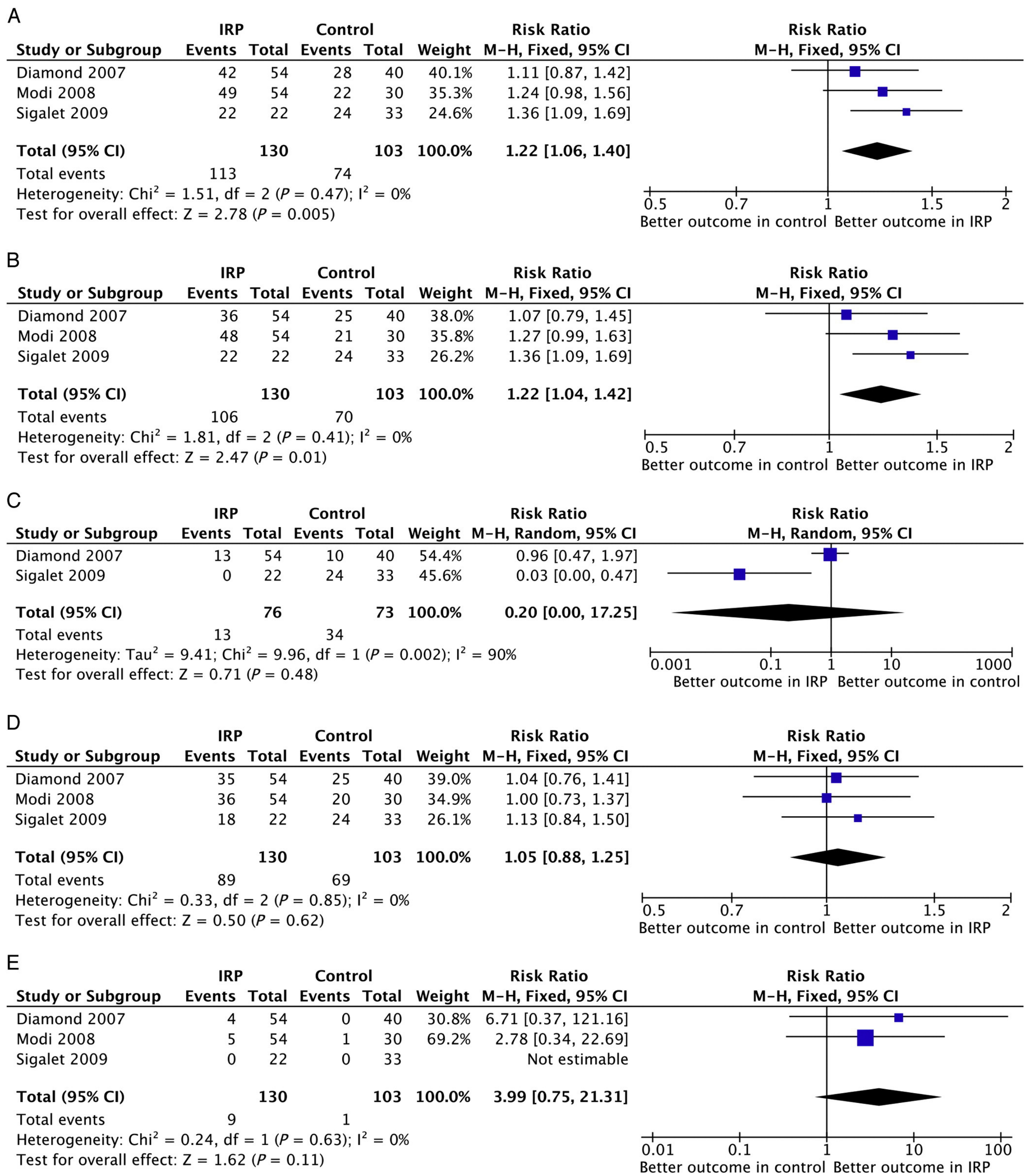

Fig. 2 A–E: Forrest Plots of Pooled Relative Risk. A, Pooled relative risk of survival from intestinal failure. B, Pooled relative risk for overall survival. C, Pooled relative risk for development of liver failure. D, Pooled relative risk for enteral autonomy. E, Pooled relative risk for transplantation.

A sensitivity analysis was performed to assess the effect of individual studies on the pooled estimate (data not shown). The disease-specific mortality was not affected by exclusion of any study. Similarly, the effect of IRP on weaning from parenteral nutrition, and the relative number of transplants was not changed by the exclusion of any study. The increase in overall survival became non-significant if the study by Sigalet et al. was excluded [4]. The RR for liver failure became significant when excluding the study by Diamond et al. ($p = 0.01$), which reflects the study results of Sigalet et al.

2.3. Summary of systematic review

2.3.1. Patient mortality

Among the various multi-disciplinary programs, patient mortality ranged from 6.4% to 37.5 % [2–4,10,12,14–20]. The wide range in mortality can in part be explained by differences in the baseline characteristics of the patient populations among the programs. Those programs reporting higher rates of mortality tended to be tertiary referral centers for large geographic areas raising the possibility of a selection bias in their results. As an example, Nucci et al. from Pittsburgh reported an overall mortality rate of 28%, however the median small bowel length at the time of referral to that program was 25 cm, which was significantly shorter than that reported in other groups [2,3,12]. The cause of death was uniform across all programs, with the two main causes being sepsis and end stage liver disease. Predictors of mortality included primary disease process, failure to achieve enteral autonomy and peak bilirubin levels [3,12,14,18]. There was also a pattern of lower mortality among the more recently published papers, which may reflect the emergence of alternative lipid management strategies in parenteral nutrition and its efficacy in mitigating cholestasis. Hess et al. followed patients over a 20 year period and reported that the only factor identified as having a significant impact on the survival of patients was the introduction of a comprehensive multi-disciplinary care team [17].

Three groups demonstrated changes in mortality when patients in multi-disciplinary programs were compared to historical controls [2–4]. All studies reported an increase in disease specific survival due to a reduction in mortality from liver failure or sepsis after the introduction of an IRP [2–4]. This increase ranged between 11% (95%-CI −13%, 42%) and 36% (95%-CI 9%, 69%) [2,4]. Modi et al. reported an increase by 24% (95%-CI −2%, 56%). The meta-analysis revealed a statistically significant RR of 1.22 (95%-CI 1.06, 1.4; $p = 0.005$). The three studies provided 233 patients for that estimate.

Overall survival increased significantly in one study [4] and non-significantly in two [2,3]. The increase ranged from 7% (95%-CI −21%, 45%) to 36% (95%-CI 9%-69%). The RR of Modi et al.'s study was 1.27 (95%-CI 0.99, 1.63). The meta-analysis showed an increase in overall survival by 22% (95%-CI 4%, 42%). All patients from the three included studies were included in this analysis.

2.3.2. Enteral autonomy

Independence from parenteral nutrition offers the best chance of survival for children with intestinal failure; therefore, enteral autonomy remains the key goal of intestinal rehabilitation programs. Rates of weaning patients from parenteral nutrition ranged from 42% to 84% across the various programs, at follow up ranging from 12 to 50 months [2–4,10,11,15,16,18]. Among those patients not able to fully wean from PN, there was a significant decrease in calories received parenterally from 100% of calories to between 32% and 56% of calories depending on the program [10,11,15].

Among those studies that compared multi-disciplinary management to historical controls, no difference was seen in the rate of enteral autonomy [2–4]. The relative risk for weaning from parenteral nutrition ranged from 1 (95%-CI 0.73, 1.37) to 1.13 (95%-CI 0.84, 1.5) [6,10]. Diamond et al. reported an RR of 1.04 (95%-CI 0.76, 1.41) [2]. The meta-analysis revealed an RR for weaning from parenteral nutrition of 1.05 (95%-CI 0.88, 1.25; $p = 0.62$). Intestinal rehabilitation programs slightly, but non-significantly, increased the risk for PN-weaning by 5%. Two hundred thirty-three patients contributed to the pooled RR. The average time to achieve enteral autonomy was variable; patients that required parenteral nutrition for longer than 2 years were less likely to achieve enteral autonomy [16].

Coordinated and multi-disciplinary management of these patients has allowed the identification of patient characteristics that predict successful weaning. Factors identified include primary diagnosis, residual gut length, early initiation of feeding and serum bilirubin [3,11,16,18]. In a multivariable analysis of factors associated with intestinal adaptation in infants with intestinal failure, Diamond et al. found that the presence of an IRP and the percent residual small bowel remaining had a positive impact on PN independence (Hazard Ratio 1.93 (95%-CI 1.14, 3.28) and 1.03 (95%-CI 1.02, 1.04) respectively), while sepsis impaired intestinal adaptation (Hazard Ratio 0.69 (95%-CI 0.60, 0.81) [28].

2.3.3. Episodes of infection

Prevention of infection is one of the main goals in the treatment of pediatric patients with intestinal failure. Sepsis has been shown to be an independent predictor of developing advanced cholestatic liver disease (OR 3.23 (95%-CI 1.8, 5.9) [31] and remains one of the leading causes of death in this population. It has been estimated that a child with intestinal failure will require 2.75 central venous catheters during their care, with an associated septic complication rate of 1.79 per 1000 catheter days [7]. Despite this, only one study has reported on the efficacy of multi-disciplinary programs in reducing the incidence of infections in these patients. Diamond et al. [2] demonstrated that patients followed by the multi-disciplinary team had fewer episodes of sepsis per month of follow up than those children in a historical cohort (0.3/mo vs 0.5/mo, respectively; $p < 0.05$). Among this group there was a trend towards a decrease in the number of gram-positive infections, however no difference in the number of gram-negative or fungal infections was observed. Recent advances in infection control include the use of rotating antibiotics to prevent bacterial overgrowth and the use of ethanol locks. Two intestinal rehabilitation programs have shown that the introduction of ethanol locks resulted in a significant decrease in the infection rate per 1000 catheter days without any adverse side effects [29,30]. The introduction of a multi-disciplinary program by Sigalet et al. resulted in a significant increase in the use of rotating antibiotics for the prevention of bacterial overgrowth, and whether this resulted in fewer episodes of infection is unclear [4].

2.3.4. Intestinal failure associated liver disease

With the advent of parenteral nutrition, long term survival in patients with intestinal failure has become a reality. However, the utilization of parenteral nutrition can result in the development of intestinal failure associated liver disease (IFALD). IFALD is a common indication for liver and/or small bowel transplantation and is one of the leading causes of death among this patient population [5,17,18]. Avoidance and/or reversal of liver dysfunction remains the key challenge to clinicians working this area and one of goals of multi-disciplinary treatment.

Strategies used to prevent and treat IFALD varied across the programs studied, and included cycling of PN, sepsis prevention, use of ursodeoxycholic acid, alternative intravenous lipid management and small bowel transplantation [2,4,10,14,15]. The treatment of patients with IFALD, has been fundamentally changed with the introduction of novel lipid management strategies such as lipid minimization or alternate lipid sources, such as omega-3 lipid emulsions.[32] Sigalet et al. and Javid et al. showed significant reductions in the serum bilirubin levels with when alternative lipid sources were utilized as part of a multi-disciplinary care approach, with levels decreasing from 112 to 8 μmol/L and from 70.1 μmol/L to 0 μmol/L [4,15]. Among those patients who had persistent evidence of IFALD management in an IRP resulted in earlier detection, shorter time to assessment for transplant and among those patients undergoing transplant improved outcomes [2,25].

The meta-analysis found liver failure decreased in the studies by Diamond et al. and Sigalet et al. [2,4]. No data were provided for that outcome by Modi et al. The risk of developing liver failure decreased by 4% (95%-CI −53%, 97%) and 97% (95%-CI 0%, -47%) in the individual studies. This decrease was statistically significant in the study by Sigalet et al. The pooled RR estimate was 0.2 (95%-CI 0, 17.25), but this decrease did not achieve statistical significance ($p = 0.48$). This estimate was calculated using data on 149 patients.

2.3.5. Surgical intervention

Pediatric patients with Intestinal Failure often undergo multiple surgical procedures. The most common autologous reconstructive procedures are those designed to repair and/or lengthen the bowel, such as the longitudinal intestinal lengthening procedure (Bianchi) and the Serial Transverse Enteroplasty (STEP). When other surgical procedures and medical management fail, patients with intestinal failure may require liver and/or small bowel transplantation.

Several programs report use of bowel lengthening procedures in order to facilitate enteral autonomy, with rates of surgery ranging from 2.6% to 63% [2,10,11,15,19,20]. The STEP was most commonly used, and in one program was associated with a median bowel length that increased from 83 to 132 cm [15]. Following bowel lengthening procedures, between 32% and 51% of patients were able to achieve enteral autonomy [10,11].

For a subset of patients with intestinal failure, transplantation becomes the only option for long term survival [33]. Rates of transplantation varied across the programs from 3% to 58%. There was variability in the rates of transplantation based on the strength of the affiliation of the program with an established transplant team. Modi et al. and Diamond et al. both showed an increase in rate of transplantation with the introduction of a multi-disciplinary team (3.3% to 9.3% and 0% to 7.5%, respectively) [5,6]. Outcomes after transplant were variable between studies, with post transplant mortality ranging from 10% to 54% [10,11,14]. In one group, 95% of patients who received a transplant and survived achieved enteral autonomy [11]. In the study be Gupte et al. the lack of care by a multi-disciplinary team at the referring hospital was associated with adverse survival among patients being assessed for and undergoing transplantation [20].

In the meta-analysis, an increase in the relative number of transplants was observed in the study by Diamond et al. and Modi et al [2,3]. This increase was 571% (95%-CI 0.37, 121.16) and 178% (95%-CI 0.34, 22.69), respectively, but it did not achieve statistical significance. The data from Sigalet et al. were not estimable, as the risk was 0 for both groups. The pooled RR was 3.99 (95%-CI 0.75, 21.31) and this was not statistically significant ($p = 0.11$). One-hundred-and-seventy-eight patients contributed to this estimate.

3. Discussion

Pediatric patients with intestinal failure are a complex patient population, who require coordinated, timely and intensive medical and surgical management. Multi-disciplinary teams are believed to be the best approach to successful and integrated care. In analysing, reflecting and reporting on their programs, several studies have reported great subjective benefit since the introduction of an IRP. In creating a National Network to track pediatric patients with intestinal failure, Guarino et al. found a significant reduction in the number of patients whose disease was of unknown origin. This has led to greater understanding of the natural history of the subsets of patients within this heterogeneous patient population [16]. Torres et al. from Nebraska also found that multi-disciplinary care permitted an increased understanding of the causes of intestinal failure [10].

Several other benefits have also been reported with multi-disciplinary IRPs, including improved monitoring of medications and nutritional strategies [3,4], more coherent, integrated planning and communication with caregivers [3,10] and more personalized patient care [11]. When adverse patient outcomes are encountered, multi-disciplinary IRPs have led to earlier detection and more timely involvement of consulting services such as transplantation assessment [2,10,15]. Coordinated care of these patients will also facilitate research endeavours, communication between institutions and care providers and ultimately the development of comprehensive clinical practice guidelines [3,15]. Although it is generally agreed that multi-disciplinary IRPs are associated with improved patient outcomes, the purpose of this study was to systematically review the current literature.

The available literature is limited to 15 articles. Twelve articles are case series and are descriptive, while 3 are analytical comparing outcomes in patients before and after implementation of an IRP. Overall, the systematic review supported the hypothesis that IRPs are associated with positive outcomes with respect to mortality, septic complications, development of IFALD, intestinal adaptation, and autologous bowel reconstruction including transplantation. The meta-analysis also showed a survival advantage to patients managed by an IRP for both disease-specific mortality (liver failure and sepsis), as well as, overall mortality.

This is the first study to summarize the current literature regarding the benefit of an IRP for pediatric patients with intestinal failure. Unfortunately, this study has many limitations that need to be addressed. First, the available literature is mainly descriptive, reflects anecdotal experience and is susceptible to all the biases of retrospective design. Second, there is tremendous heterogeneity between the studies in several areas:

i) The study populations differ depending on the size of the program and degree of tertiary or quaternary referral;
ii) The individual IRP programs themselves possess variable team membership, infrastructure, and experience;
iii) The nature of the IRP “exposure” is not consistent. Based on our analysis the treatment approach by each program is not standardized and reflects in most cases the institutional culture. In spite of differing treatment protocols, the IRP itself was considered ‘the intervention’ for the purpose of the meta-analysis. Any treatment provided under the umbrella of an established IRP was part of that intervention;
iv) The variable definitions for clinical outcomes make comparison between studies difficult. For instance, there was no standardized definition of short bowel syndrome to define the population. In addition, there were no standardized definitions for reporting cholestasis, advanced IFALD, liver failure, septic rates. Even residual bowel length was often reported in centimetres where it should be reported as the percentage of expected gut length for gestational age [34];
v) The studies comprise different time eras where treatment options were variable; One would expect articles published more recently to exhibit better outcomes as contemporary therapies became utilized, and
vi) The relatively shorter follow-up in IRP patients may impact clinical outcome evaluation.

As a result of the limitations identified in the present literature, there is an opportunity to improve future work. We would propose the adoption of standardized definitions for short bowel syndrome, and specific clinical outcomes in this population, and would encourage authors to report them in a consistent manner. This would facilitate the comparison of results between programs. Better yet would be the creation of a multi-institution intestinal failure registry to evaluate therapies and compare outcomes as individual center sample sizes are small.

Now that survival has been improved in many IRP centres, there are an increasing number of younger, more anatomically complex patients who are free of advanced IFALD and at risk of chronic complications. Therefore, the focus of research now needs to shift to evaluation of other long-term morbidities such as metabolic bone disease, and neuro-developmental outcomes. We should be examining issues related to quality of life (patient as well as care providers), as well as, cost effectiveness.

The management of infants and children with intestinal failure can be expensive and place heavy demands on health care resources. Olieman et al. [13] from the Netherlands estimated the level of resource consumption in the management of infantile short bowel syndrome. They described the costs associated with the interdisciplinary management of 10 patients over a period of 4 years. In their series, the median length of initial hospital admission was 174 days, with a median length of stay of 409 days at the end of follow up. Patients had on average 5 surgical interventions. They estimated the direct patient care costs in this group to be €268,700 per patient, and that the cost escalated with patient age. Within their population, the associated costs were higher among those patients that could not be weaned from parenteral nutrition. In a similar study, Spencer et al. [35] determined in 41 pediatric short bowel syndrome patients, that the mean total 5 year hospital and homecare expenditure for each patient was $619,851 ± $1,028,985 (US dollars). No study to date has compared the cost of multi-disciplinary IRP care to the costs associated with intestinal transplantation. There has also been no economic evaluation of IRP that also considers quality of life or utility in addition to cost alone.

The management of infants and children with intestinal failure is undergoing an exciting transition, where advances in nutritional support, sepsis control and surgical technique are making long term survival a reality. However, optimizing these elements of care is a complex task and needs to be provided in a coordinated, multi-disciplinary fashion. The institution of multi-disciplinary care teams has demonstrated improvement in patient outcomes, with improved enteral autonomy, reduced infectious complications, reduced incidence of IFALD and most importantly, clearly documented improvements in survival. As the field of intestinal rehabilitation moves forward, multi-disciplinary care will provide the ideal platform for further research including the evaluation of new treatment strategies, cost-effectiveness and quality of life, the development of national patient data registries and comprehensive clinical practice guidelines.

References

[1] Duro D, Kamin D, Duggan C. Overview of pediatric short bowel syndrome. J Pediatr Gastroenterol Nutr 2008;47:S33-6.

[2] Diamond IR, de Silva N, Pencharz PB, et al. Neonatal short bowel syndrome outcomes after the establishment of the first Canadian multidisciplinary intestinal rehabilitation program: preliminary experience. J Pediatr Surg 2007;42:806-11.

[3] Modi BP, Langer M, Avery Ching Y, et al. Improved survival in a multidisciplinary short bowel syndrome program. J Pediatr Surg 2008;43:20-4.

[4] Sigalet D, Boctor D, Robertson M, et al. Improved outcome in paediatric intestinal failure with aggressive prevention of liver disease. Eur J Pediatr Surg 2009;19:348-53.

[5] Lloyd DAJ, Vega R, Bassett P, et al. Survival and dependence on home parenteral nutrition: experience over a 25-year period in a UK referral centre. Aliment Pharmacol Ther 2006;24:1231-40.

[6] Quiros-Tejeira RE, Ament ME, Reyen L, et al. Long-term parenteral nutritional support and intestinal adaptation in children with short bowel syndrome: a 25 year experience. J Pediatr 2004;145:157-63.

[7] Gandullia P, Lugani F, Costabello L, et al. Long-term home parenteral nutrition in children with chronic intestinal failure: a 15 year experience at a single Italian centre. Dig Liver Dis 2011;43:28-33.

[8] Fishbein TM, Schiano T, LeLeiko N, et al. An integrated approach to intestinal failure: results of a new program with total parenteral nutrition, bowel rehabilitation, and transplantation. J Gastrointest Surg 2002;6:554-62.

[9] Sudan D, DiBaise J, Torres C, et al. A multidisciplinary approach to the treatment of intestinal failure. J Gastrointest Surg 2005;9:165-77.

[10] Torres C, Sudan D, Vanderhoof J, et al. Role of an intestinal rehabilitation program in the treatment of advanced intestinal failure. J Pediatr Gastroenterol Nutr 2007;45:204-12.

[11] Koehler AN, Yaworski JA, Gardner M, et al. Coordinated interdisciplinary management of pediatric intestinal failure: a 2-year review. J Pediatr Surg 2000;35:380-5.

[12] Nucci A, Cartland Burns R, Armah T, et al. Interdisciplinary management of pediatric intestinal failure: a 10-year review of rehabilitation and transplantation. J Gastrointest Surg 2008;12:429-36.

[13] Olieman JF, Poley MJ, Gischler SJ, et al. Interdisciplinary management of infantile short bowel syndrome: resource consumption, growth and nutrition. J Pediatr Surg 2010;45:490-8.

[14] Cowles RA, Ventura KA, Martinez M, et al. Reversal of intestinal failure-associated liver disease in infants and children on parenteral nutrition: experience with 93 patients at a referral center for intestinal rehabilitation. J Pediatr Surg 2010;45:84-8.

[15] Javid PJ, Malone FR, Reyes J, et al. The experience of a regional pediatric intestinal failure program: successful outcomes from intestinal rehabilitation. Am J Surg 2010;199:676-9.

[16] Guarino A, De Marco G. and the Italian National Network for Pediatric Intestinal Failure. Natural history of intestinal failure, investigated through a national network-based approach. J Pediatr Gastroenterol Nutr 2003;37:136-41.

[17] Hess RA, Welch KB, Brown PI, et al. Survival outcomes of pediatric intestinal failure patients: analysis of factors contributing to improved survival over the past two decades. J Surg Res 2011;170:27-31.

[18] Colomb V, Dabbas-Tyan M, Taupin P, et al. Long term outcome of children receiving home parenteral nutrition: a 20-year single-center experience in 302 patients. J Pediatr Gastroenterol Nutr 2007;44:347-53.

[19] Khalil BA, Ba'ath ME, Aziz A, et al. Intestinal rehabilitation and bowel reconstructive surgery: improved outcomes in children with short bowel syndrome. J Pediatr Gastroenterol Nutr 2012;54:505-9.

[20] Gupte GL, Beath SV, Protheroe S, et al. Improved outcomes of referrals for intestinal transplantation in the UK. Arch Dis Child 2007;92:147-52.

[21] Moher D, Liberati A, Tetzlaff J, et al. Preferred reporting items for systematic reviews and meta-analyses: the PRISMA statement. Int J Surg 2010;8:336-41.

[22] Liberati A, Altman DG, Tetzlaff J, et al. The PRISMA statement for reporting systematic reviews and meta-analyses of studies that evaluate healthcare interventions: explanation and elaboration. BMJ 2009;339: b2700.

[23] Wells GA, Shea B, O'Connell D, et al. The Newcastle–Ottawa Scale (NOS) for assessing the quality if nonrandomized studies in meta-analyses. http://www.ohri.ca/programs/clinical_epidemiology/oxford.htm. [accessed on 2010 Nov 11].

[24] DerSimonian R, Kacker R. Random-effects model for meta-analysis of clinical trials: an update. Contemp Clin Trials 2007;28:105-14.

[25] Higgins J, Thompson S, Deeks J, et al. Statistical heterogeneity in systematic reviews of clinical trials: a critical appraisal of guidelines and practice. J Health Serv Res Policy 2002;7:51-61.

[26] Higgins JP, Thompson SG. Quantifying heterogeneity in a meta-analysis. Stat Med 2002;21:1539-58.

[27] DerSimonian R, Laird N. Meta-analysis in clinical trials. Control Clin Trials 1986;7:177-88.

[28] Diamond IR, Struijs M-C, deSilva N, et al. Does the colon play a role in intestinal adaptation in infants with short bowel syndrome? a multivariable analysis. J Pediatr Surg 2010;45:975-9.

[29] Jones BA, Hull MA, Richardson DS, et al. Efficacy of ethanol locks in reducing central venous catheter infections in pediatric patients with intestinal failure. J Pediatr Surg 2010;45:1287-93.

[30] Wales PW, Kosar C, Carricato M, et al. Ethanol lock therapy to reduce the incidence of catheter-related bloodstream infections in home parenteral nutrition patients with intestinal failure: preliminary experience. J Pediatr Surg 2011;46:951-6.

[31] Diamond IR, de Silva NT, Tomlinson GA, et al. The role of parenteral lipids in the development of advanced intestinal failure associated liver disease in infants: a multivariable analysis. JPEN J Parenter Enteral Nutr 2011;35:596-602.

[32] Diamond IR, Sterescu A, Pencharz PB, et al. Changing the paradigm: Omegaven for the treatment of liver failure in pediatric short bowel syndrome. J Pediatr Gastroenterol Nutr 2009;48:209-15.

[33] Shatnawei A, Parekh N, Rhoda KM, et al. Intestinal failure management at the Cleveland clinic. Arch Surg 2010;145:521-7.

[34] Wales PW, Christison-Lagay E. Short bowel syndrome: epidemiology and etiology. Semin Pediatr Surg 2010;19:3-9.

[35] Spencer AU, Kovacevich D, McKinney-Barnett M, et al. Pediatric short bowel syndrome: the cost of comprehensive care. Am J Clin Nutr 2008;88:1552-9.